

<Capstone Title>

A Capstone Project by

**<Last Name, First Name Middle Initial>
<Last Name, First Name Middle Initial>
<Last Name, First Name Middle Initial>
<Last Name, First Name Middle Initial>
<Last Name, First Name Middle Initial>**

**Submitted to the Faculty of Information Technology
Bulacan State University-Sarmiento Campus**

**In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

<Month year>

CHAPTER I

INTRODUCTION

This chapter presents the general overview of the study. It introduces the context of the project, identifies the problem to be addressed, and states the objectives, significance, scope, and key terms of the study. This chapter serves as the foundation of the capstone project and guides the reader in understanding the purpose and direction of the research.

Background of the Study

(start your discussion of your introduction here) The Background of the Study provides the contextual foundation of the capstone project. It explains the existing situation, problem, or gap in a specific domain that the proposed system aims to address. This section should begin with a broad discussion of the topic and gradually narrow down to the specific issue being studied.

The background should describe the current systems, practices, or workflows, identify limitations or inefficiencies in the existing setup, and emphasize the relevance and necessity of the proposed information technology solution. Claims and observations must be supported by credible sources using APA citation style.

The final paragraph of this section must contain the Statement of the Problem, clearly stating the core issue or gap that the project intends to address.

Objectives of the Study

Begin the general objective with “To” or “The general objective of this study is to...”

The specific objectives follow, using a colon at the end:

1. Use action verbs (e.g., design, develop, implement, evaluate, analyze).
2. Each specific objective must be clear, concise, and measurable.
3. Avoid compound objectives (one objective = one action).
4. Ensure a direct correspondence between each objective and the conclusions in Chapter V.
5. When system evaluation is included, the objectives must explicitly state the evaluation standard, such as ISO/IEC 25010.
6. There must be a one-to-one correspondence between: specific objectives → evaluation results (Chapter IV), and conclusions/recommendations (Chapter V).
7. Use sub-objectives only when a specific objective requires further breakdown for clarity.

Significance of the Study

The Significance of the Study explains the importance, value, and expected impact of the project. This section identifies the stakeholders who will benefit from the system, such as students, faculty, administrators, the institution, and future researchers.

It should clearly describe how the system will benefit each group by improving efficiency, accessibility, accuracy, or decision-making. The discussion must focus on practical and academic contributions of the study.

Scope and Delimitations

The **scope** outlines the extent of the study or system, including its key features, functions, intended users, and supported platforms (e.g., web, mobile). It also describes the specific processes the system will address, improve, or automate. The scope should be closely aligned with the objectives of the study, ensuring that the project remains focused, feasible, and achievable within the given timeframe and resources.

Limitations refer to the weaknesses or constraints in the research design, data, tools, or methods that are beyond the researchers' control. These may include factors such as time constraints, limited data availability, dependency on third-party systems, or the unpredictability of user input. Identifying these limitations helps contextualize the results and ensures transparency regarding the study's potential constraints.

The scope and limitations should together present a balanced view of what the study will accomplish, while acknowledging the practical barriers that may influence its outcomes.

Definition of Terms

This section provides operational definitions of technical, domain-specific, or project-related terms used in the study. Terms should be arranged

alphabetically and defined based on how they are used within the context of the project. Only essential terms should be included to ensure clarity and consistency for the readers.

Word 1

Definition for Word 1

Word 2

Definition for Word 2

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents an overview of previous research, relevant concepts, and existing systems that are directly related to the proposed capstone project. It serves as a foundation for understanding the technological and conceptual background of the study, identifying gaps in current systems, and justifying the need for the proposed solution.

Review of Related Literature

Summarize scholarly works, articles, journals, and previous capstone projects relevant to the topic. Discuss theories, methodologies, and findings that support the project's design. Use **APA citation style** for all sources and avoid plagiarism.

(You may add subsections to your related literature discussion to organize the review according to specific topics.)

Review of Related Systems

Identify at least **6–10 existing systems** similar to your proposed project. For each system, provide the **name**, **developer (if known)**, **main features**, **target users**, and **technologies used**. Discuss the **strengths**, **limitations**, and what differentiates your proposed system from the other.

(You may add subsections in the discussion of related systems to organize them based on their functionalities, architectures, or technologies used.)

Comparison of Related Systems

After describing each system in narrative form, include a **comparison table** that visually summarizes and contrasts its key features.

For tables, use the following format:

Table 1. From "References" > Insert Caption > Table

Indent the Table label to start of table	The List of Tables should be automatically generated
--	--

Synthesis

The Synthesis serves as the summary and interpretation of the reviewed literature and systems. It highlights the key themes, patterns, gaps, and opportunities identified during the review. It also reinforces how the current study builds upon or differs from past work.

CHAPTER III

DESIGN AND METHODOLOGY

Requirement Analysis and Documentation

In this section, describe the requirements for your system.

Identify and clearly describe the **functional requirements** (what the system must do) and **non-functional requirements** (performance, usability, reliability, etc.).

Explain how requirements were gathered. Mention the tools or methods used, such as **interviews**, **surveys**, **observations**, or **document analysis**.

If applicable, include a summary table of requirements.

Conceptual Framework

Present the conceptual framework as a **single diagram** that shows the relationships between major components of the study. Make sure the figure is centered on the page.

(Insert  Figure)

After the diagram, provide a narrative that explains how the elements relate to one another and how the framework supports the development of your system.

Development Methodology

Describe the software development model used in the project (e.g., **Agile**, **Waterfall**, **Rapid Application Development**).

- Choose one SDLC: Agile, Waterfall, or RAD — and explain why you chose it

- Add CRISP-DM for the analytics component
- Make a phases table with columns:
 - o Phase Name
 - o Agile SDLC Activities
 - o CRISP-DM Activities
 - o Key Deliverable

Why CRISP-DM? It is the standard process for data mining and machine learning projects. It covers: Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment. This maps directly to how your algorithm was developed..

System Design

Present design diagrams that explain the overall structure and behavior of the system. Each diagram should be numbered, labeled, and followed by a clear explanation.

Context Diagram (Proposed)

Show how your system interacts with external entities (users, external systems, databases).

Data Flow Diagram – Level 1

Illustrate the flow of data through the system's major processes.

Use Case Diagram

Depict user roles and their interactions with the system functions.

Entity Relationship Diagram (ERD)

Show database entities, attributes, and relationships.

System Architecture Diagram

Present the structure of the system components, including how analytics modules or data processing components fit within the architecture.

Flowchart

Provide flowcharts for major processes, logic decisions, or algorithms.

Mockups**Tools, Technologies, and Development Environment**

List and describe the technologies used to build and test the system, such as:

- Programming languages (e.g., JavaScript, Dart)
- Frameworks and libraries (e.g., React, Flutter, Chart.js)
- Backend technologies (e.g., Node.js, PHP)
- Databases (e.g., MongoDB, MySQL)
- APIs (e.g., Google Maps, Analytics APIs)
- Development environments and testing tools (e.g., VS Code, Postman)

Explain why these tools were chosen and how they support system development.

Development Cost

Identify and describe the major functional modules of the system. Use subsections or bullet points for clarity.

For each module, describe what it does and how users interact with it.

Evaluation Methods

This section describes how the system **will be evaluated** following implementation. The objective of the evaluation is to assess the **technical quality** of the system and its **acceptability to users**.

To achieve this, two established evaluation models will be used:

- **ISO/IEC 25010 Software Product Quality Model**
- **Technology Acceptance Model (TAM)**

You may adapt the structure of this section based on your evaluation instruments or sampling plans.

Evaluation Using ISO/IEC 25010

Describe how the system's technical quality will be evaluated.

- **Purpose:** To assess software quality using international standards.
- **Quality Characteristics:** Functional Suitability, Usability, Performance Efficiency, Reliability, Maintainability, Portability.
- **Respondents:** A minimum of **three (3) IT professionals** with software/system expertise will be invited as evaluators.
- **Instrument:** An evaluation form based on ISO 25010 criteria will be prepared. A **5-point Likert scale** (1 = Strongly Disagree to 5 = Strongly Agree) will be used for each item.
- **Process Overview:** Explain that responses will later be tabulated and analyzed in Chapter IV.

Evaluation Using Technology Acceptance Model (TAM)

Explain how user acceptance will be assessed.

- **Purpose:** To measure user perceptions of usefulness, ease of use, and intention to adopt the system.
- **TAM Constructs:** Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Behavioral Intention to Use (BI).
- **Respondents:** A group of **15 to 30 end users** who have interacted with the system will be selected.
- **Sampling Method: Purposive sampling** will be used to ensure respondents have first-hand experience.
- **Instrument:** A TAM-based questionnaire with a **5-point Likert scale** will be used to collect responses.

Data Handling and Analysis

This is the most structured subsection. Follow this exact order:

A. Respondents and Population Description

State the total population (N) and break it down by role. Identify the two evaluator groups:

1. End-user respondents for TAM — 15 to 30, purposive sampling
2. IT professional evaluators for ISO/IEC 25010 — minimum 3, purposive sampling

B. Data Analysis

Write these three things in order:

1. Describe the 5-point Likert scale:

5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, 1 – Strongly Disagree

2. Insert the **Likert Scale Conversion Table**:

Rating Range	Interpretation
4.51 – 5.00	Highly Acceptable
3.51 – 4.50	Acceptable
2.51 – 3.50	Moderately Acceptable
1.51 – 2.50	Slightly Acceptable
1.00 – 1.50	Not Acceptable

3. Write a narrative paragraph describing what each range means for your specific system.

C. Statistical Treatment of Data

State that the **Arithmetic Mean (AM)** is used. Present the formula:

$$AM = \Sigma(w_i f_i) / n$$

Define each variable:

- AM = Arithmetic Mean
- w_i = Likert scale weight (5, 4, 3, 2, 1)
- f_i = Frequency of responses for each weight
- n = Total number of respondents

Explain that AM is computed per item, per quality characteristic, per TAM construct, and for the overall score.

D. Statistical Tools Used

Present three tools in this order:

Frequency — used to count how many respondents chose each rating per item.

Percentage — used to express the proportion of responses. Formula:

$$P = (F \times 100) / N$$

Where: P = Percentage, F = Frequency, N = Total respondents

Weighted Mean — connects back to the AM formula. This is the main tool for interpreting results.

E. Algorithm Performance Metrics

The performance of the embedded algorithm will be evaluated independently from the system quality assessment conducted under ISO/IEC 25010 and the user acceptance evaluation conducted under TAM. While the ISO/IEC 25010 and TAM evaluations measure the quality and acceptability of the web-based system as a whole, the algorithm performance evaluation specifically measures how accurately and reliably the analytical model produces its outputs. The metrics used depend on the type of algorithm integrated into the system.

REFERENCES

Books

Author, A. A. (Year). *Title of the book: Subtitle if any*. Publisher.

Example:

Shelly, G. B., & Rosenblatt, H. J. (2011). *Systems analysis and design* (9th ed.). Course Technology.

Journal Articles

Author, A. A., & Author, B. B. (Year). Title of the article. *Title of the Journal*, Volume number(Issue number), page range. <https://doi.org/xxxx>

Example:

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

Websites

Author or Organization. (Year, Month Day). Title of the webpage. *Website Name*. URL

Example:

Commission on Higher Education. (2022, March 15). *CHED Annual Report 2021*. <https://ched.gov.ph/annualreport2021/>

Thesis or Capstone Project

Author, A. A. (Year). *Title of the thesis or capstone project* (Unpublished undergraduate thesis). Name of Institution.

Example:

Reyes, J. C. (2023). *A mobile-based scheduling system for university laboratories* (Unpublished undergraduate thesis). Polytechnic University of the Philippines.